Deep Learning **Spring-2026**

Term Project

The Term Project is an opportunity for you to apply what you have learned in class to a problem of your interest. Potential projects usually fall into these two tracks:

- **Applications**: If you're coming to the class with a specific background and interests (e.g. biology, engineering, physics), you can apply DeepNets to problems related to your particular domain of interest. Pick a real-world problem and apply DeepNets to solve it.

  This type of projects would involve careful data preparation, an appropriate loss function, details of training and cross-validation and good test set evaluations and model comparisons.

- **Models:** You can build a new model (algorithm) with DeepNets, or a new variant of existing models, and apply it to tackle vision/speech/NLP tasks. This track might be more challenging, and sometimes leads to a piece of publishable work.

  For models, DeepNets have been successfully used in a variety of learning tasks. This type of projects would involve understanding the state-of-the-art models, and building new models or improving existing models for a given task.

You can find some sample projects from the following links:

https://cs231n.stanford.edu/2024/reports.html

https://cs231n.stanford.edu/2022/reports.html

Project Proposal

Your project proposal should be between 2 – 3 pages. The following is a suggested structure for your report:

> **Title, Author(s)**
> **Introduction**: this section introduces your problem, and the overall plan for approaching your problem
> **Problem statement**: Describe your problem, precisely specifying the dataset to be used, expected results and evaluation
> **Technical Approach**: Describe the methods you intend to apply to solve the given problem

Please submit your proposal as a PDF on LMS. Only one person on your team should submit.

Final Report

Your final write-up is required to be no more than 15 pages (single column). The following is a suggested structure for your report:

> **Title, Author(s)**
> **Abstract:** Briefly describe your problem, approach, and key results. Should be no more than 300 words.
> **Introduction**: Describe the problem you are working on, why it's important, and an overview of your results

Related Work: Discuss published work that relates to your project. How is your approach similar or different from others?
Data: Describe the data you are working with for your project. What type of data is it? Where did it come from? How much data are you working with? Did you have to do any preprocessing, filtering, or other special treatment to use this data in your project?
Methods: Discuss your approach for solving the problems that you set up in the introduction. Why is your approach the right thing to do? Did you consider alternative approaches? You should demonstrate that you have applied ideas and skills built up during the semester to tackling your problem of choice. It may be helpful to include figures, diagrams, or tables to describe your method or compare it with other methods.
Experiments: Discuss the experiments that you performed to demonstrate that your approach solves the problem. The exact experiments will vary depending on the project, but you might compare with previously published methods, perform an ablation study to determine the impact of various components of your system, experiment with different hyperparameters or architectural choices, use visualization techniques to gain insight into how your model works, discuss common failure modes of your model, etc. You should include graphs, tables, or other figures to illustrate your experimental results.
Conclusion: Summarize your key results - what have you learned? Suggest ideas for future extensions or new applications of your ideas.

Supplementary Material: not counted toward your 15 page limit and submitted as a separate file. Your supplementary material might include:
Source code (if your project proposed an algorithm, or code that is relevant and important for your project.).
Cool videos, interactive visualizations, demos, etc.

Submission: You will submit your final report as a PDF and your supplementary material as a separate PDF or ZIP file through LMS.

Additional Submission Requirements:
Any code that was used as a base for projects must be referenced and cited in the body of the report. This includes any assignment code, finetuning example code, open-source, or Github implementations. You can use a footnote or full reference/bibliography entry.

Collaboration Policy:
You can work in teams of 2 or 3 people.

Project Presentation:
Groups will present their project during the final exam slot at the end of semester. Each presentation should be no more than 10 mins.

Important Dates
Project proposal: due Friday, March 27.
Final report: due Monday, June 1.
Presentation session: Wednesday, June 3 (9:00-12:00).

APPENDIX:

You can check out the most recent work in Deep Learning from the following web page: https://paperswithcode.com/

The list below presents some papers on important historical advances of DeepNets in the computer vision community:

Image Classification: [Krizhevsky et al.], [Russakovsky et al.], [Szegedy et al.], [Simonyan et al.], [He et al.], [Huang et al.], [Hu et al.] [Zoph et al.]
Object detection: [Girshick et al.], [Ren et al.], [He et al.]
Image segmentation: [Long et al.] [Noh et al.] [Chen et al.]
Face recognition: [Taigman et al.] [Schroff et al.] [Parkhi et al.]
Depth estimation: [Eigen et al.]
Image-to-sentence generation: [Karpathy and Fei-Fei], [Donahue et al.], [Vinyals et al.] [Xu et al.] [Johnson et al.]

You might also gain inspiration by taking a look at some popular computer vision datasets:

Meta Pointer: A large collection organized by CV Datasets.
ImageNet: a large-scale image dataset for visual recognition organized by WordNet hierarchy
SUN Database: a benchmark for scene recognition and object detection with annotated scene categories and segmented objects
Places Database: a scene-centric database with 205 scene categories and 2.5 millions of labelled images
NYU Depth Dataset v2: a RGB-D dataset of segmented indoor scenes
Microsoft COCO: a new benchmark for image recognition, segmentation and captioning
Flickr100M: 100 million creative commons Flickr images
Labeled Faces in the Wild: a dataset of 13,000 labeled face photographs
Human Pose Dataset: a benchmark for articulated human pose estimation
YouTube Faces DB: a face video dataset for unconstrained face recognition in videos
UCF101: an action recognition data set of realistic action videos with 101 action categories
HMDB-51: a large human motion dataset of 51 action classes
ActivityNet: A large-scale video dataset for human activity understanding
Moments in Time: A dataset of one million 3-second videos

You can find source codes and model files of popular DeepNets from GitHub pages or Model Zoos. Some examples are given below:

TensorFlow Models: https://github.com/tensorflow/models
Model Zoo: https://modelzoo.co/frameworks

Detectron: [He et al.] https://github.com/facebookresearch/Detectron
Tracking Progress in NLP: https://github.com/sebastianruder/NLP-progress